

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING**

|                  |                                                                                                                                               |               |                                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|
| Course Code:     | CSA0305                                                                                                                                       | Course Title: | Data Structures                                |
| CO Assessed:     | CO2 – Manipulation (BL3, BL4)                                                                                                                 | Assessment:   | Assessment Tool 1 – Analytical Problem Solving |
| Weightage:       | 30% of CIA                                                                                                                                    | Total Marks:  | 100                                            |
| CO2 Statement:   | Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques. |               |                                                |
| Student Name:    | J. MEHER KOUSHIK                                                                                                                              | Reg. No.:     | 192525211                                      |
| Section / Batch: | AI&ML                                                                                                                                         | Date:         | 20/07/2026                                     |

**Code of Conduct**

I J. MEHER KOUSHIK (192525211) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: J. MEHER KOUSHIK

**Instructions**

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

| Section / Topic                              | Focus                           | Q Nos            |
|----------------------------------------------|---------------------------------|------------------|
| Linked Data Structure, Search Data Structure | List Operations, Binary Search  | 1                |
| Stack Data Structure, Queue Data Structures  | Stack Notations, Priority Queue | 2                |
| <b>GRAND TOTAL:</b>                          |                                 | <b>100 Marks</b> |

192525192

### Annexure A – Analytical Problem Solving

#### ANALYTICAL PROBLEM SOLVING

| S.No | REG NO    | NAME           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 192472062 | ANTHOTI ROHITH | <ol style="list-style-type: none"> <li>1. Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(10), Pop(), Push(44), Push(33), Pop(), Pop(), Pop(). Print the top element of the data structure and its index position. With a neat sketch explained in detail.</li> <li>2. How do you implement Queue data structure using Singly Linked List? With a neat diagram discussing Enqueue and Dequeue operations using SLL.</li> </ol>                                                                                                                                                                           |
|      | 192572047 | AARTHI SREE G  | <ol style="list-style-type: none"> <li>1. Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail.</li> <li>2. Implement Binary searching Algorithm for searching an element and its position having the following 10 elements. Elements are 66, 111, 212, 323, 432, 543, 643, 649, 777, 888. Case 1: Key element = 111, Case 2: Key element = 888, Case 3: Key element = 999. Analyse the concept in detail.</li> </ol> |

|    |           |                      |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----|-----------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8  | 192511397 | H SARAVANA           | <ol style="list-style-type: none"> <li>1. Compare the performance of a linear queue when 100 enqueue and dequeue performed alternately.</li> <li>2. A queue implemented using an array report "Overflow" though some positions are empty. <ol style="list-style-type: none"> <li>a. Analyze why this occurs.</li> <li>b. Explain how a circular queue solves this problem.</li> </ol> </li> </ol>                             |
| 9  | 192525345 | INTURI JASWANTHI     | <ol style="list-style-type: none"> <li>1. Consider the singly linked list: <math>10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50</math>. Delete the node containing 30. Explain how the list changes after deletion.</li> <li>2. Given two sorted linked lists:<br/> L1: <math>1 \rightarrow 3 \rightarrow 5 \rightarrow 7</math><br/> L2: <math>2 \rightarrow 4 \rightarrow 6 \rightarrow 8</math></li> </ol> |
| 10 | 192525211 | JUJARU MEHER KOUSHIK | <ol style="list-style-type: none"> <li>1. Convert the following infix expression into postfix using a stack. Show the contents of the stack after processing each symbol. <math>(A+B)*(C-D)/E</math></li> <li>2. Convert the following infix expression into postfix using operator precedence and associativity influence the conversion. <math>A+B*C-(D/E+F)*G</math></li> </ol>                                            |

### Rubrics for Calculation

| Criteria                 | Weightage | Level 4 - Exemplary                                                                        | Level 3 - Proficient                                        | Level 2 - Developing                                    | Level 1 - Beginning                      |
|--------------------------|-----------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------|------------------------------------------|
| Problem Understanding    | 20        | Demonstrates complete and precise understanding; identifies all constraints and edge cases | Shows clear understanding with minor gaps                   | Partial understanding; misses some constraints          | Misinterprets problem or lacks clarity   |
| Data Structure Selection | 20        | Selects optimal data structure with strong justification and comparison                    | Selects appropriate structure with reasonable justification | Selects somewhat suitable structure with weak reasoning | Incorrect or no data structure selection |
| Algorithm Design         | 20        | Designs efficient, logically sound, and wellstructured algorithm                           | Algorithm is correct with minor inefficiencies              | Algorithm is partially correct or lacks clarity         | Algorithm is incorrect or missing        |
| Accuracy of Solution     | 30        | Error-free, wellstructured, optimized code/solution                                        | Minor errors but overall functional                         | Multiple errors; partially working                      | Non-functional or missing solution       |

|                        |    |                                                               |                                                  |                                      |                              |
|------------------------|----|---------------------------------------------------------------|--------------------------------------------------|--------------------------------------|------------------------------|
| Clarity                | 10 | Exceptionally clear, wellorganized, uses diagrams effectively | Clear and organized with <del>minor</del> issues | Somewhat organized but lacks clarity | Poorly organized and unclear |
| Faculty In-charge      |    | Course Coordinator                                            |                                                  | Program Director                     |                              |
| Signature: <u>Ugbo</u> |    | Signature: _____                                              |                                                  | Signature: _____                     |                              |
| Date: _____            |    | Date: _____                                                   |                                                  | Date: _____                          |                              |

convert the following infix expression into postfix notation using stack. show the contents of the stack after processing each symbol  $(A+B)*(C-D)/E$

A) Infix Expression:  $(A+B)*(C-D)/E$

| Symbol | Stack | Postfix          |
|--------|-------|------------------|
| (      | (     |                  |
| A      | (     | A                |
| +      | ( +   | AB               |
| B      | ( +   | AB +             |
| )      | -     | AB +             |
| *      | *     | AB + C           |
| (      | *( -  | AB + C D         |
| C      | *( -  | AB + C D -       |
| -      | *( -  | AB + C D -       |
| D      | *( -  | AB + C D -       |
| )      | *     | AB + C D -       |
| /      | /     | AB + C D - *     |
| E      | /     | AB + C D - * E   |
| END    | -     | AB + C D - * E / |

Final Post Fix Expression:

$AB + CD - *EI$

- 2) convert the following infix Expression into Post fix Form. Explain how operator Precedence and associativity influence the conversion

$A + B * C - (D / E + F) * G$

Q3) Infix Expression:

$A + B * C - (D / E + F) * G$

step-by step-conversion

| Symbol | Stack | Postfix |
|--------|-------|---------|
| A      | -     | A       |
| +      | +     | A       |
| B      | +     | AB      |
| *      | +*    | AB      |
| C      | +*    | ABC     |
| -      | -     | ABC*+   |
| (      | -(    | ABC*+   |
| D      | -(    | ABC*+D  |

|     |       |                          |
|-----|-------|--------------------------|
| /   | - ( / | $ABC^* + C$              |
| E   | - ( / | $ABC^* + DE$             |
| +   | - ( + | $ABC^* + DE  $           |
| F   | - ( + | $ABC^* + DE   F$         |
| )   | -     | $ABC^* + DE   F +$       |
| *   | - *   | $ABC^* + DE   F +$       |
| G   | - *   | $ABC^* + DE   F + G$     |
| End | -     | $ABC^* + DE   F + G^* -$ |

Final Postfix Form:-

$$ABC^* + DE | F + G^* -$$

Effect of Precedence and Associativity

- 1) Parentheses ( ) have the highest Priority, so their contents are processed first
- 2) Multiplication \* and division / have higher Precedence than addition + and subtraction -.

3)

3) In  $B * C$ ,  $*$  is performed before  $+$ , giving  $B * C$ .

4) In  $D / E + F$ ,  $/$  is performed before  $+$ , giving  $D / E + F$ .

5)  $*$  and  $/$  have the same precedence and are Left-associative, so they are processed from left to right.

*Ujjwal*